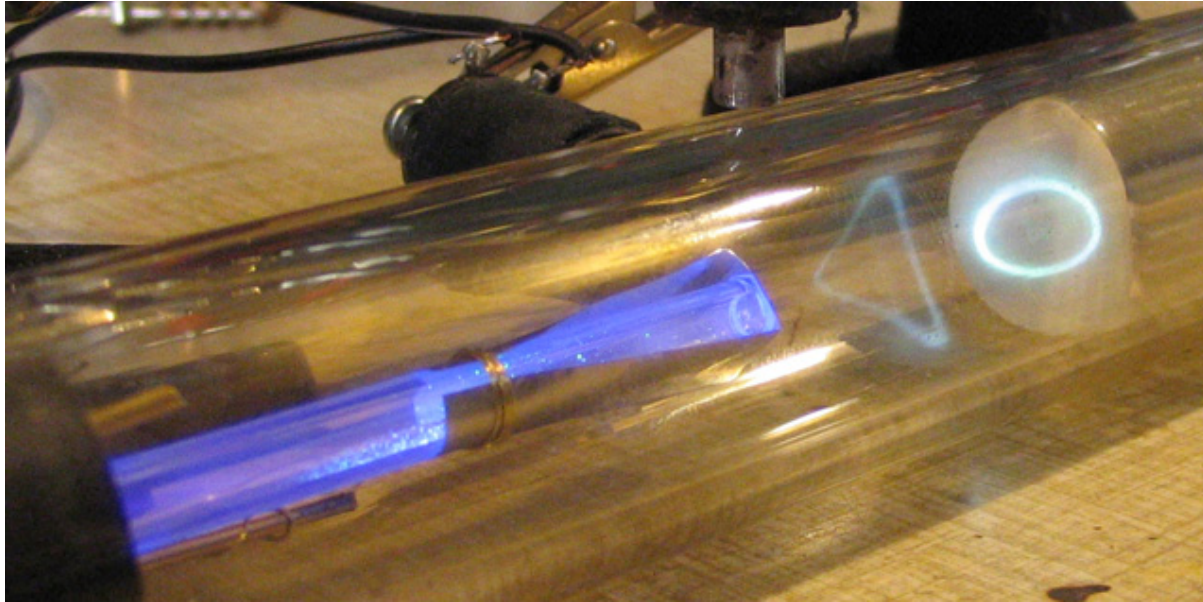


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Updated May 2012.

Welcome to sparkbangbuzz.com.

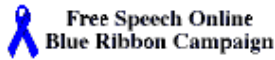
This is a homepage of really fun scientific and technical projects that are easy to build and informative.

Disclaimer

Anyone reading this web page should not assume that any given subject is safe or legal. The purpose here is to give information only. I take no responsibility for what anyone may do with the information given here. What you do with it is your own business and responsibility. Building some of the projects, described here, can be dangerous, illegal or both; and require that the builder or user be very conscientious and able to exercise a great deal of Caution and Common Sense. A project described as "relatively safe" may not be completely safe. It may be described as "relatively safe" only because it appears to be safer than making a batch of Nitroglycerine. Never get cocky where safety is

concerned.

POLITICAL CORRECTNESS!!! This page welcomes all varieties of race, gender or anyone interested in its contents. If you are offended by anything that seems politically incorrect, don't write to me. I do not care to be annoyed by fanatic whiners who expect every printed line to pass correct political specifications.



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Zinc Negative Resistance Oscillator.

Anyone can make an active semiconductor device at home by heating galvanized sheet metal. With it, simple rf and audio oscillators can be built. Amplifiers can also be built. With a carbon microphone or transformer connected, the rf oscillator can even transmit audio to a nearby am radio. Really!! I'm not kidding.

[Zinc Negative Resistance Oscillator](#)

Zinc Negative Resistance Crystal Oscillators and Zinc 80 Meter CW Transmitter.

[Zinc Negative Resistance Crystal Oscillators.](#)

This is a continuation of my first article on zinc negative resistance oscillators made from the simple homemade zinc negative resistance diode.

In the early 1920's Oleg Losev in Russia experimented with negative resistance oscillators and circuits using zincite. One of the main problems was that usable zincite came from only one place in the world - New Jersey.

My articles are about doing very similar experiments using heat treated galvanized sheet metal. I believe Losev would have been very excited if he had been able to try heat treated galvanized sheet metal as a substitute for zincite and it may very well work even better.

Zinc Negative Resistance Makes 40 Meter CW Transmitter.

[Zinc Negative Resistance Makes 40 Meter Transmitter.](#)

A homemade semiconductor is used to make a 40 meter CW transmitter.

Zinc Negative Resistance 20 Meter CW Transmitter.

[Zinc Negative Resistance 20 Meter Transmitter.](#)

A homemade semiconductor can even make a 20 meter CW transmitter.

Homemade Zinc Oxide Detector Diode.

[Zinc Oxide Detector Diode.](#)

Unlike the zinc negative resistance diodes used to make oscillators and transmitters, this diode is a rectifying detector diode.

Homemade Memristor.

[Homemade Memristor.](#)

A homemade memristor was made from pieces of brass, copper or lead that had been turned to a dark color from exposure to sulfur.

Homemade Copper Oxide Thermoelectric Generator Can Light An LED.

[Homemade Copper Oxide Thermoelectric Generator.](#)

Simple homemade copper oxide thermoelectric generator can be made from two pieces of copper wire. A single junction can generate more than 300 millivolts when heated with a flame. Sixteen of these in series can light an LED.

Thermocouple Made From Ordinary Copper And Steel Wire.

[Copper And Steel Thermocouple.](#)

A thermocouple made of ordinary copper and steel wire can generate enough current through a coil to deflect a compass needle using just the heat from my fingers.

Fly and Levitate Objects With Static Electricity.

[Static Electricity Flyers.](#)

Light objects made from plastic grocery bag material can be flown and levitated above a charged balloon or piece of styrofoam. A simple electrostatic glider can be made that flies just like a walkalong glider.

Evaporograph.

[Evaporograph.](#)

This simple to make evaporograph will show images of cold objects when they make contact with a sensitive membrane. The membrane can also be made to show images of infrared heat.

Homemade Magnetic Amplifiers using common 12 Volt Transformers.

[Homemade Magnetic Amplifiers using common 12 Volt Transformers.](#)

This article is to de-mystify the esoteric magnetic Amplifier and show how one can be built at home using common everyday 12 volt transformers.

Homemade Magnetic Audio Amplifier using toroids found in parts box.

[Homemade Magnetic Audio Amplifier.](#)

An audio amplifier can be built without using any tubes or transistors.

Homemade FET Transistor Made From Cadmium Sulfide Photoresistor.

[FET Transistor Homemade From Cadmium Sulfide Photocell.](#)

A common cadmium photoresistor (called photocell in the text) was made into a crude insulated gate field effect transistor. Definite transistor action was observed after improvising a gate on the photocell. Even though the voltage gain was small, a considerable amount of power gain was produced.

Corona Oscillator And Corona Triode.

[Corona Oscillator And Corona Triode.](#)

A pin point near a flat positive high voltage plate forms an oscillator that produces pulses roughly in the AM broadcast band frequency range. A control element near the pin point can control the overall current and frequency. This makes a triode that exhibits gain.

"Easy Ten" A simple 80 Meter CW Transmitter Using A 2N3904.

A very simple 80 meter CW one transistor QRP transmitter made from a common npn transistor, two resistors, two capacitors, a 3.5 - 4 mhz crystal and a 9v battery. I call this transmitter "The Easy Ten" because it can be easily heard from a distance of over ten miles.

This article includes some good basic philosophy on antennas.

["Easy Ten" A simple 80 meter CW Transmitter Using A 2N3904.](#)

Flame Triode With Gain.

I have found a simple way to make a triode with gain using an alcohol flame. This flame

triode exhibits both power gain and voltage gain. By making a flame dual triode, I have been able to make a free running multivibrator oscillator and code practice oscillator.

I also observed a very intriguing phenomenon of flame electrical conductivity; that electrons from a cathode, placed inside of the flame, can flow to the anode even when the anode appears to be placed well outside of the flame.

[Flame Triode With Gain.](#)

Homemade Cathode Ray Tubes.

[Homemade Cathode Ray Tubes.](#)

Homemade T.E.A. Lasers.

[Homemade T.E.A. Lasers.](#)

A simple and crude conversion of a lawnmower to run on propane. Runs Great!!.

[Propane Powered Lawnmower.](#)

Experiments with a borax or baking soda rectifier. [borax rectifier.](#)

A simple homemade wide range variable electrolytic capacitor using baking soda can vary in capacitance in one continuous sweep over a 5000 to 1 range. This capacitor can vary the frequency of a simple relaxation oscillator over a similar range. Even the zinc negative resistance oscillator can be continuously swept over the entire audio range.

[Homemade wide range variable electrolytic capacitor and oscillator.](#)

Copper Oxide is really neat stuff. It is very easy to make a simple photocell capable of deflecting a volt meter with a small flashlight or listening to audio from a sound modulated light beam. [Very Simple Homemade Photocell](#)

It is easy to make a thermistor, thermoelectric generator or pressure sensor by heating a piece of copper wire. [Homemade Thermistor and Pressure Sensor](#)

A drop of salt water on some aluminum can produce some very interesting electronic sounds when amplified. [Very interesting sounds from a drop of salt water on aluminum.](#)

Do you get more thrust by blowing air out a tube than you would get in the negative direction by sucking the same amount of air into the tube??? I did some experiments and found out a pretty definitive answer. The answer may be surprising to many. [Jet Negative Thrust by Sucking???](#)

I have always been fascinated with the old Arc transmitters of the early period of wireless radio. Not to be confused with Spark Gap transmitters, the Arc transmitter was different and was supposed to produce a continuous wave signal that could be sound

modulated. What follows is a description of a crude circuit that I built. [Simple Arc Transmitter](#)

The coherer was one of the first detectors of radio signals ever used 100 years ago. I tried building one. Getting impressive results turns out to be much easier than I have always thought. [Simple Homemade Coherer](#)

It is incredibly easy to make a homemade device that behaves similar to a tunnel diode. Using it, a simple rf oscillator was made which could be coaxed into running at 12 mhz. [Homemade Tunnel Diode and RF Oscillator.](#)

I made an active semiconductor device at home with Iron Pyrites. With it, a simple continuous wave broadcast band oscillator (transmitter) was built. With a carbon microphone connected, it could even transmit my voice to a nearby am radio. [Iron Pyrites Negative Resistance Oscillator](#)

More ways to make a homemade negative resistance devices that are much superior to iron pyrites in ease of use and consistency. [More Negative Resistance Materials.](#)

A simple home made negative resistance audio code practice oscillator that also makes experimenting with negative resistance materials fun and easy without having to use an oscilloscope or curve tracer. [Negative Resistance Code Practice Oscillator.](#)

Zinc negative resistance RF amplifier for crystal sets and negative resistance regenerative receivers. [Zinc RF Amplifier](#)

I have always wondered if a vacuum tube could operate with a degree of vacuum attainable by amateur means. This is how I broke open a vacuum tube triode and operated it after pumping it down with a vacuum pump. [Home Evacuated Vacuum Tube](#)

The next step - an actual homemade vacuum tube diode. [Home Made Vacuum Diode](#)

The next step - first attempt at making an actual homemade vacuum tube triode. [Home Made Vacuum Triode](#)

Magnetic levitation using permanent magnets. No superconductors, no energy input for stabilization and no spinning tops. Earnshaw - take a hike. [Permenant Magnet Levitation.](#)

Magnet Kicker keeps any magnet in motion to make simple motors or "perpetual motion" toys. [Magnet Kicker.](#)

Can ordinary propane, such as that used for fuel, be used as a refrigerant? It sure can!!

[Homebuilt Refrigeration System from hardware store parts.](#)

The next step. Can an old refrigerator operate when charged with ordinary propane? [Old Refrigerator Charged With Propane.](#)

Throw staples, tacks, screws etc. like a knife thrower and stick them in the target. Fun and impressive hobby. [How to throw staples, tacks, screws etc.](#)

Almost everyone gets a kick from making a good sounding explosion, especially if it can be done with very easy to get materials. Well here is a relatively safe way of doing it with compressed air. [Airbang](#)

A "Dry Ice Bomb"??? Who Needs Dry Ice? [Airbomb](#)

Simple Static Electricity Generator. The Electrophorus is wimpy, the Wimshurst Machine performs but is much more difficult to build. This [Simple Static Electricity Generator.](#) is incredibly simple to make and fills in part of the huge performance gap that exists between the Electrophorus and the Wimshurst Machine.

Send sound on a modulated LED light beam. [Sound Modulated Light Beam.](#)

Send sound on a modulated Laser Beam. [Sound Modulated Laser.](#)

Send sound on a Modulated Flashlight Beam. [Sound Modulated Flashlight.](#)

Broadcast a signal through your entire house with a magnetic field. [Broadcast with magnetism.](#)

Thirty foot long steel wire makes far out music. [Wiremusic.](#)

Links

Videos of sparkbangbuzz.com projects and other web interesting web sites.

[Links.](#)